Adopting ELT: Who Transforms What?

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2023-02-13
Halifax dbt Meetup Q1

Overview

- Characterizing ETL & ELT
  - A Process of Two Teams
  - In Three Tasks
- Criteria for Sharing the Work
  - Two Examples

From ETL to ELT

Extract

- Gather Data

Transform

- Shape & Calculate

Load

- Deliver Data

Extract

- Gather Data

Load

- Deliver Data

Transform

- Shape & Calculate

Two Teams

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Dev

- Deep infrastructure knowledge
- Robust version control
- Fixed sprints & prioritization

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Analyst

- Deep domain knowledge
- Customer needs
- BI tool familiarity
- Short, flexible “sprint” lengths

ETL: Three Tasks

Extract

- Gather from disparate sources

- Stability
- Efficiency

Transform

- Create the schema
- Calculate attributes
- Calculate metrics

- Domain knowledge
- Customer knowledge
- Rapid prototyping

Load

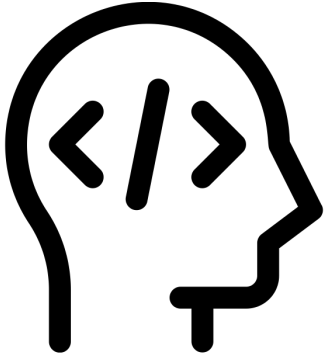
- Make visible to analysts

- Stability
- Efficiency

ELT: Three Tasks in a Better Order

Extract

- Stability
- Efficiency



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Load

- Stability
- Efficiency

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Transform

- Domain knowledge
- Customer knowledge
- Flexibility

Should analysts do all transformation?

No

Sharing the work: team strengths

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Dev

- More efficient transformation tools
- But less time for wild experimentation

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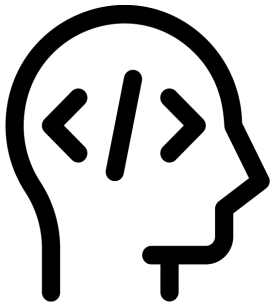
Analyst

- Better placed to iterate on definitions and schema design
- But limited tools for implementation

Example: calculating new clients and patients

- Our customers are veterinary clinics
- Two common metrics
  - New clients (humans)
  - New patients (pets)
- Internal question: who should calculate these metrics?



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Example: transaction data

TRANSACTION_DATETIME	CLIENT_ID	PATIENT_ID	ITEM_ID	REVENUE_CATEGORY_NAME
2023-06-28 10:15:06.000	19231-52093	116514-52093	54560-52093	Euthanasia
2020-07-10 00:00:00.000	34658-52093	86088-52093	RTG-52093	Diagnosis and Other
2021-04-06 13:32:28.000	37977-52093	94365-52093	1825-52093	Surgery
2020-08-10 16:07:14.000	18640-52093	79104-52093	2523-52093	Bravecto
2023-05-05 16:39:30.000	30640-52093	93330-52093	51-52093	Exam Services
2023-03-28 11:16:07.000	20827-52093	66452-52093	2487-52093	Rehabilitation
2020-06-01 13:14:33.000	36114-52093	94393-52093	28023-52093	Nexgard Spectra
2021-09-27 08:20:15.000	30984-52093	104874-52093	RTG-52093	Diagnosis and Other
2022-11-23 14:36:46.000	35006-52093	94464-52093	3-52093	Exams
2023-06-22 16:20:35.000	46167-52093	116386-52093	1-52093	Exams
2020-12-16 10:15:56.000	39399-52093	97595-52093	2215-52093	Surgery
2019-04-29 14:51:41.000	32388-52093	80545-52093	3261-52093	Vaccines

- Connect to Practice Information Management Software (PIMS)
- Get line item data from every invoice (transactions fact table)
- Augment the data with categorization

Example: client and patient data

CLIENT_ID	FIRST_NAME	LAST_NAME	PIMS_ENTERED_DATE	EARLIEST_TRANSACTION_DATE	CREATED_AT_UTC
19231-52093	xxx	xxx	2009-06-01	2019-05-14	2023-03-22 20:07:09.169
29230-52093	xxx	xxx	2015-11-02	[NULL]	2023-03-22 20:07:09.169
42264-52093	xxx	xxx	2021-10-15	[NULL]	2023-03-22 20:07:09.169
45514-52093	xxx	xxx	2023-03-09	2023-03-14	2023-03-22 20:07:09.169

PATIENT_ID	NAME	PIMS_ENTERED_DATE	BIRTH_DATE	DEATH_DATE	EUTHANASIA_DATE	EARLIEST_MEDICAL_SERVICE_DATE
26936-52093	xxx	2007-10-02	1994-06-14	2010-05-08	2010-05-08	[NULL]
116514-52093	xxx	2023-06-28	2023-06-14	2023-06-29	2023-06-28	2023-06-28
77610-52093	xxx	2017-02-22	2016-06-22	[NULL]	[NULL]	[NULL]
112207-52093	xxx	2022-10-13	2021-10-13	[NULL]	[NULL]	2022-10-27

- Client and patient data (clients and patients dim tables)
  - a bunch of personal information
  - entered date
- Calculate a few attributes
  - first_transaction_date
  - euthanasia_date

Example: New Clients

CLIENT_ID	FIRST_NAME	LAST_NAME	PIMS_ENTERED_DATE	EARLIEST_TRANSACTION_DATE	CREATED_AT_UTC
19231-52093	xxx	xxx	2009-06-01	2019-05-14	2023-03-22 20:07:09.169
29230-52093	xxx	xxx	2015-11-02	[NULL]	2023-03-22 20:07:09.169
42264-52093	xxx	xxx	2021-10-15	[NULL]	2023-03-22 20:07:09.169
45514-52093	xxx	xxx	2023-03-09	2023-03-14	2023-03-22 20:07:09.169

Definition

If the entered date is within the transaction history, use the first transaction date. Otherwise, use the entered date.

Widely Accepted

- All our customers like this definition
- Used across multiple products
- Haven't changed it in ages

Requires Only Raw Data

- clients.pims_entered_date
- min(transaction_date) group by server
- min(transaction_date) group by client

How heavy?

- server-wide earliest transaction is expensive for large servers

Example: New Patients

PATIENT_ID	NAME	PIMS_ENTERED_DATE	BIRTH_DATE	DEATH_DATE	EUTHANASIA_DATE	EARLIEST_MEDICAL_SERVICE_DATE
26936-52093	xxx	2007-10-02	1994-06-14	2010-05-08	2010-05-08	[NULL]
116514-52093	xxx	2023-06-28	2023-06-14	2023-06-29	2023-06-28	2023-06-28
77610-52093	xxx	2017-02-22	2016-06-22	[NULL]	[NULL]	[NULL]
112207-52093	xxx	2022-10-13	2021-10-13	[NULL]	[NULL]	2022-10-27

Definition

A patient can't be new and deceased on the same day. Aside from that, if the patient's entered date is outside the transaction history, use that. Otherwise, use the first medical service.

Widely Customized

- Some customers count deceased
- Some use first medical service, others any transaction
- Currently considering a change to first transaction

Requires Augmented Data

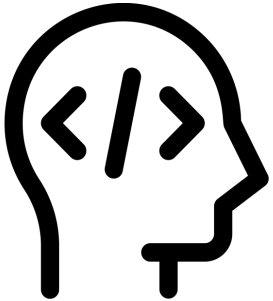
- euthanasia date & medical service require categorization
- that categorization must be downstream from dev load process

How heavy?

- similar to clients
- shares some components
  - server first transaction

Conclusion: Who Transforms What?

New Clients



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- Efficiency
  - a little heavy for SQL
- Definition
  - stable
  - broadly accepted
  - requires only raw data

New Patients

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- Efficiency
  - pretty heavy for SQL
- Definition
  - open questions
  - customization
  - requires augmented data

Breakout Discussion

What transformations would you like to take control of?

What tools do you have to inform this decision?